Data Structures Lecture 1: Tree Architectures

Binary Search Trees and Self Balancing AVL Nodes

A binary search tree enforces the invariant that left subtree keys are strictly smaller than root and right are greater. Unbalanced binary trees degrade to linear O(N) linked list complexity in the worst-case ascending insertion order. AVL trees maintain a balance factor of minus one, zero, or one using single and double rotation operations.

Data Structures Lecture 2: Graph Exploration

Depth-First Search and Breadth-First Shortest Paths

Graphs represent pairwise node relationships through vertices connected by directed or undirected weighted edges.
Breadth-first search traverses level by level using a FIFO queue to discover shortest paths in unweighted graphs.
Depth-first search uses a recursion stack to explore branches deeply and compute topological sorts in DAGs.

Data Structures Lecture 3: Dynamic Programming

Optimal Substructure and Overlapping Subproblems

Dynamic programming solves complex optimization problems by decomposing them into memoized subproblem sol
Top-down memoization caches recursive results in a lookup table, whereas bottom-up tabulation builds iteratively.
The classic 0-1 knapsack problem establishes a two-dimensional recurrence relation bounded by total weight capac